

Bitwise differential trail through **QALQAN** found by SAT

CiVerLy

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1 QALQAN

Maximal differential probability: 2^{-38}

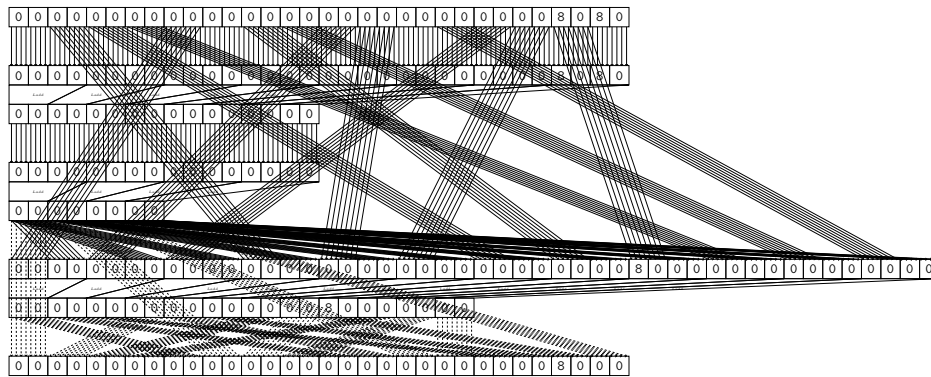
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
KcapAddStart																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
B Bus Layer																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0	
L																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	
QuaternBurst																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
QuaternBurst																																	
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	0	0	0	8	0	0	0	0	0	8	0	8	0	
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	0	0	0	8	0	0	0	0	0	8	0	8	0	

2 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

3 L

Maximal differential probability: $2^{-0.0}$

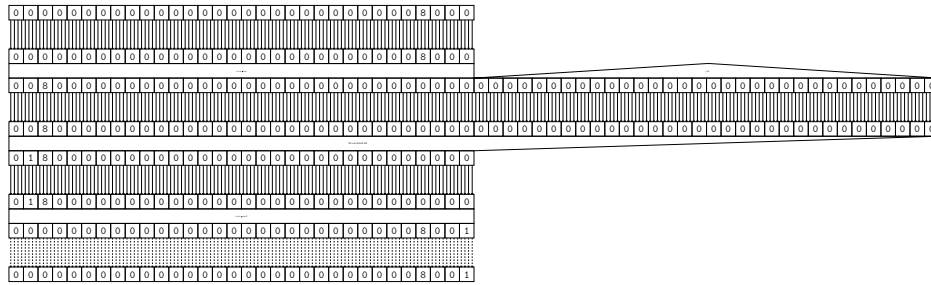


4 QalqanRound

Maximal differential probability: $2^{-18.0}$ [illegible]

5 Add128

Maximal differential probability: $2^{-2.0}$

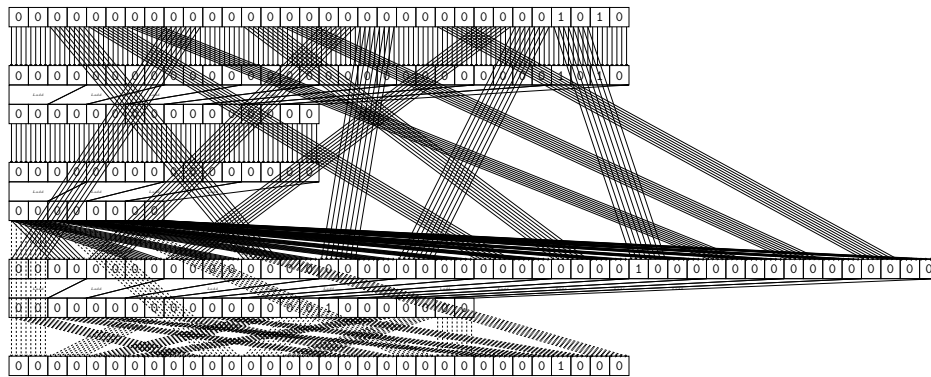


6 SBoxLayer

Maximal differential probability: $2^{-14.0}$ [illegible]

7 L

Maximal differential probability: $2^{-2.0}$

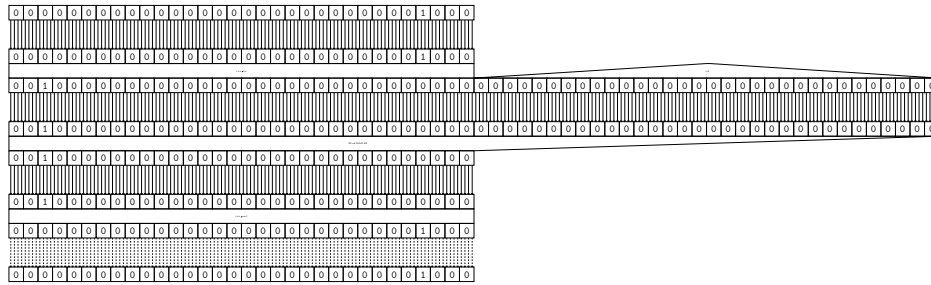


8 QalqanRound

Maximal differential probability: $2^{-8.0}$

[illegible]

9 Add128

Maximal differential probability: $2^{-1.0}$ 

10 SBoxLayer

Maximal differential probability: $2^{-7.0}$

[illegible]

11 L

Maximal differential probability: $2^{-0.0}$

